

Forecasting EV Registration Growth in New York State: A Controlled Comparison of Five Methods Under a Trend Reversal

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Abstract

Utilization data for individual EV fast-charging stations is commercially licensed and not publicly available, so this project asks a narrower question: using only public New York State DMV and charging-infrastructure data, can county-level EV registration growth be forecast reliably enough to be useful for site selection. We build a reproducible pipeline over 210,774 registered electric vehicles and 5,501 public charging stations, then compare five forecasting methods – a naive last-year baseline, an OLS linear trend, Holt linear exponential smoothing, Random Forest, and LightGBM – on a 2024–2025 holdout. Every method more sophisticated than the naive baseline loses to it, and the size of the loss (naive RMSE 481; OLS 686; Random Forest 1,175; LightGBM 1,526; Holt 2,246) tracks almost exactly how strongly each method’s inductive bias assumes the trend continues. The reason is a real regime change: statewide registrations grew every year from 2018 through 2023, then declined in both 2024 and 2025. We decompose each method’s error into bias and variance components, show LightGBM’s forecast is driven almost entirely by one- and two-year lag features, and use the resulting zip-level charging-station data to flag 129 New York City zip codes with 50 or more registered EVs and zero DC fast chargers.

1 Introduction

Electric vehicle charging site selection depends on a forecast of future demand, and the data that would forecast it most directly – per-station session counts, dwell time, up-time – is sold by vendors such as Paren and Atlas Public Policy rather than published. This project builds the closest available substitute from public data: New York State DMV vehicle registration snapshots and a state-maintained directory of public charging stations. Rather than assume a single model is the right one to trust, we hold everything constant except the forecasting method – same counties, same hold-out years, same features where applicable – and compare five methods that make progressively stronger assumptions about trend continuation.

Our questions are simple: which method wins on a real holdout, is the winner an artifact of an easy holdout window, and what does the losing methods’ failure actually look like once decomposed. The answer to the second question turns out to be the most useful finding in the project.

2 Data

Two live New York State open-data sources and one federal geography file are pulled and cached by `src/ingest.py`:

- **NY DMV Vehicle Registrations** (`data.ny.gov, w4pv-hbkt`): filtered to `fuel_type=ELECTRIC` and `record_type=VEH`, giving 210,774 electric vehicles across 62 counties and 1,907 zip codes, model years 2011–2026.
- **NY EV Charging Stations** (`data.ny.gov, 7rrd-248n`): 5,501 public charging locations with Level 1, Level 2, and DC fast port counts per zip.

- **Census Gazetteer ZCTA file**: land area per zip code, used for density rather than population, which is not available without an API key.

Model year is used as a proxy for adoption timing: it is not a perfect stand-in for the calendar year a vehicle was first registered, but it is the only year field the DMV snapshot exposes, and it is stated as a limitation throughout rather than treated as exact. County-year counts are built by grouping registrations by county and model year, reindexing every county across the full 2011–2025 year range so that a county with zero new registrations in a given year is recorded as zero rather than missing. Two lag features (one and two years back) and a year index are derived from this series for the tree-based and naive methods.

3 Methods

All five methods are trained on model years 2011–2023 and evaluated on a strictly held-out 2024–2025 window per county, with the final partial year (2026) excluded from both training and evaluation entirely.

Naive predicts next year’s count as equal to last year’s – no trend assumption at all. **OLS trend** fits one ordinary least squares line per county across the full training window. **Holt linear** (Holt, 1957) fits an exponentially-smoothed level and trend per county and extrapolates the trend forward with no bound. **Random Forest** (Breiman, 2001) and **LightGBM** (Ke et al., 2017) are both trained on four features – year index, a county indicator, and the two lag features – pooled across all counties, with 300 trees (depth 6) and 200 boosting rounds (depth 4, learning rate 0.05) respectively.

For each method we report RMSE, MAE, MAPE, and a

bias-variance decomposition of the holdout error: bias is the mean signed error, and the variance component is defined so that the two recombine exactly to RMSE:

$$\text{variance component} = \sqrt{\max(\text{RMSE}^2 - \text{bias}^2, 0)} \quad (1)$$

4 Results

4.1 Main comparison

Table 1 and Figure 1 report holdout results. The naive baseline wins outright: RMSE 481 versus 686 for OLS trend, 1,175 for Random Forest, 1,526 for LightGBM, and 2,246 for Holt linear.

Table 1: Holdout results, model years 2024–2025.

Model	RMSE	MAE	Bias
Naive (last year)	481	162	+84
OLS trend	686	264	-236
Random Forest	1,175	601	+514
LightGBM	1,526	730	+669
Holt linear	2,246	759	+757

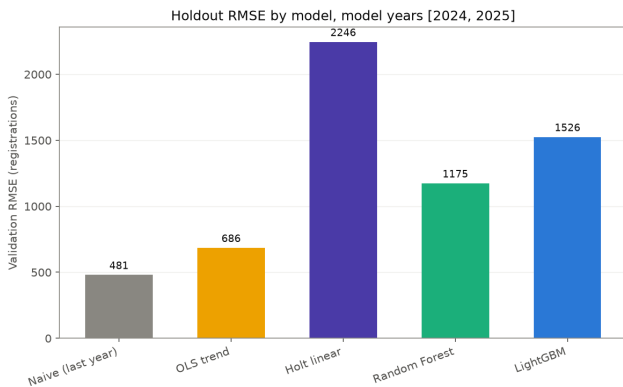


Figure 1: Holdout RMSE by model. The ordering tracks how strongly each method extrapolates trend, not general model quality.

4.2 Why the naive baseline wins

Figure 2 shows why: statewide new registrations rose from 11,795 (2021) to a peak of 48,411 (2023), then fell to 44,396 (2024) and 38,046 (2025) – two consecutive years of decline after five years of acceleration. Every method except naive was trained on a window that ends in uninterrupted growth, and each extrapolates that growth by an amount proportional to how directly its structure encodes a trend. Holt linear extrapolates an explicit, unbounded trend term and has both the worst RMSE (2,246) and the largest bias (+757). Random Forest and LightGBM split on lag features, which

bounds their extrapolation to values seen in training, producing smaller but still large positive bias (+514 and +669). OLS is the only non-naive method with negative bias (-236): a single line fit across the entire 2011–2023 window is pulled down by the early, near-flat years enough to partially offset the late acceleration it is also trying to fit.

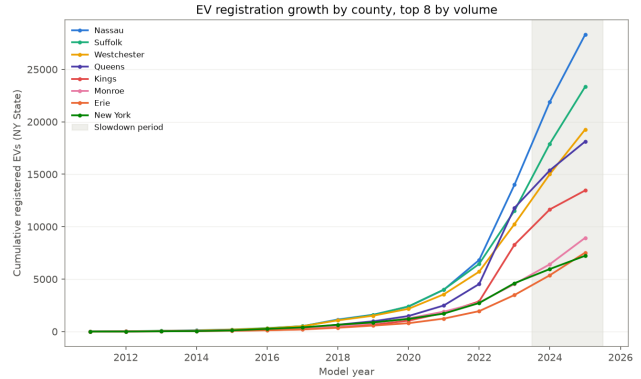


Figure 2: Cumulative registrations for the top 8 counties by volume. All show the same flattening after 2023 (shaded region).

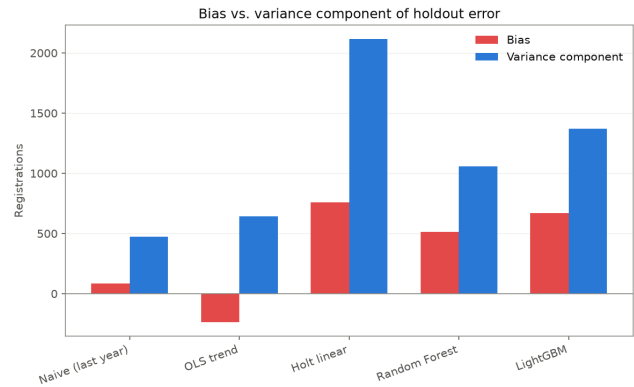


Figure 3: Bias and variance components of holdout RMSE. Every non-naive method is bias-dominated in the same direction: over-prediction.

4.3 Where LightGBM fails and what drives it

Figure 4 breaks LightGBM’s holdout error down by county. New York (Manhattan) county has the single worst error (RMSE 4,716) and it is almost entirely bias (4,716) rather than variance (40) – on a county with only two holdout points, the model was confidently and consistently wrong in one direction, which is as much a small-sample warning as a modeling one. Orange and Rockland counties split their error more evenly between bias and variance, suggesting a noisier underlying trend rather than a single systematic miss.

Figure 5 shows why LightGBM behaves this way: by total gain, the one- and two-year lag features dominate the

model, with year index and the county indicator contributing much less. LightGBM is, in effect, a flexible way of looking at recent history, which is exactly why it is hurt by the same regime change that hurts Holt: recent history stopped being a reliable guide to what came next.

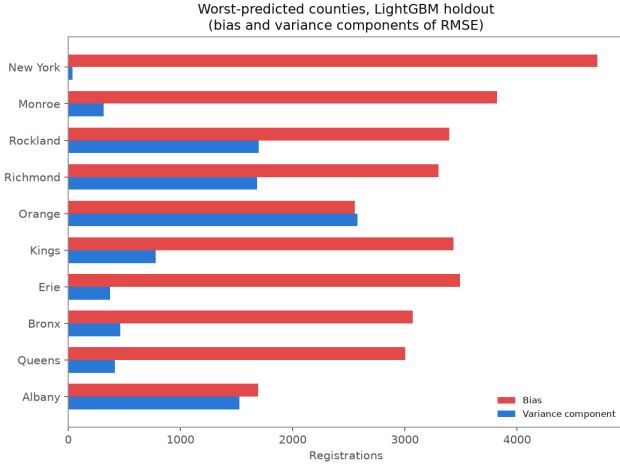


Figure 4: Bias and variance components of LightGBM holdout error, worst 10 counties.

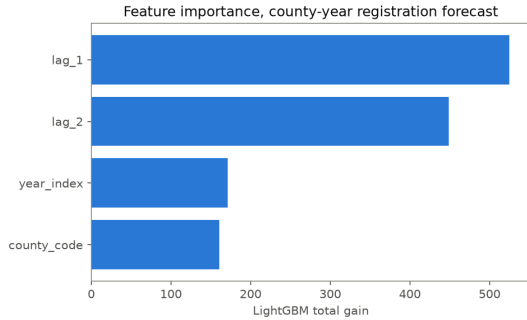


Figure 5: LightGBM feature importance by total gain. Lag features dominate.

4.4 Application: where NYC charging supply lags registered demand

Independent of the forecasting comparison, the same cleaned registration and station data can answer a simpler question: where is the gap between registered EVs and existing charging supply largest today. Figure 6 ranks New York City zip codes with at least 20 registered EVs by registered EVs per public charging port. Restricting to zip codes with 50 or more registered EVs, 129 zip codes across the five NYC counties have zero DC fast charging ports at all.

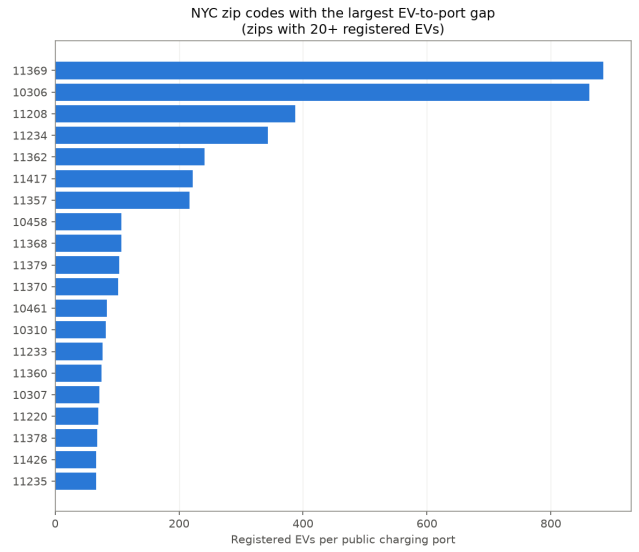


Figure 6: NYC zip codes with the largest ratio of registered EVs to existing charging ports.

5 Discussion

The headline result is not that naive forecasting is a good strategy in general – it is that a holdout period which happens to break the training trend will make every trend-aware method look worse than a method with no opinion about trend at all, and that this is worth reporting plainly rather than picking a friendlier holdout window. A production forecasting system for this problem would need either an explicit regime-change detector or wide enough uncertainty bands to survive being wrong about one, and neither is something a single point-accuracy metric on a lucky holdout window would reveal.

6 Conclusion

Across five forecasting methods evaluated identically on the same New York State EV registration data, holdout accuracy ordering tracked each method’s willingness to extrapolate a training-period trend, and the least sophisticated method won because the trend it would have extrapolated had already reversed. The bias-variance decomposition shows this clearly: every method beyond naive is bias-dominated in the same direction, and the size of that bias scales with how directly the method encodes recent trend. Separately, the same pipeline surfaces 129 New York City zip codes with real registered EV demand and no DC fast charging access today, which does not depend on any forecast being correct.

Code and Data

All code, tests, and result files are available at github.com/Akhilesh-Vangala/ev-charging-demand-forecasting.

References

- [1] L. Breiman, “Random forests,” *Machine Learning*, vol. 45, no. 1, pp. 5–32, 2001.
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- [6] Paren, “State of the Industry: DC Fast Charging,” paren.app, 2026.